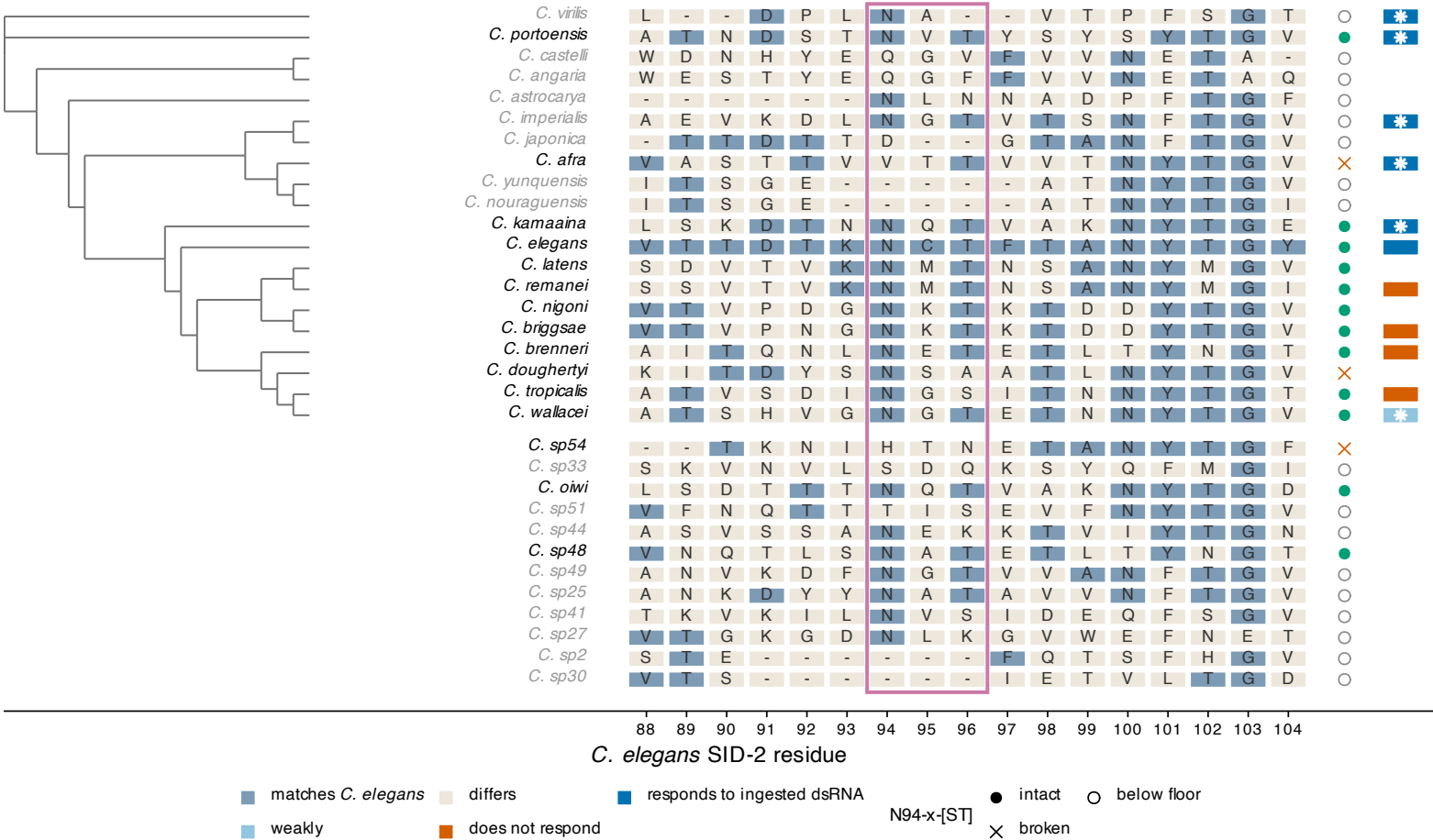


A Residue 96 is conserved, and does not predict the phenotype

Residues aligned to *C. elegans* 88-104, ordered on the Open Tree of Life topology (cladogram, left; no branch lengths). The boxed columns are the N94-x-T96 sequon; the rightmost column is the published response to ingested dsRNA. Residue 96 is Thr in 11 of the 14 readable species, Ser in 1 and Lys in none – yet of the 9 with a published call, 5 respond and 4 do not, and the sequon does not separate them: *C. briggsae*, *C. remanei*, *C. brenneri* and *C. tropicalis* carry the intact motif and are insensitive. *C. afra*, which has lost Asn94 and is a natural AxT, responds. Dimmed names are below the block-identity floor and counted neither way; an asterisk marks a phenotype scored on a different isolate from the sequenced one; species with no position in the reference tree are listed below it.



B Removing the conserved sequon does not phenocopy 96K

N94A removes the sequon while leaving residue 96 alone, so it should phenocopy 96K if the glycan is what 96K destroys. It does not: AxT has no sequon and is fully resistant, while NxK in the same background costs 42 points. It is the lysine, not the missing glycan – which is what panel A's residue-96 column independently implies. Wilson 95% intervals; p is Fisher's exact test against the wild type of the same background, with an arrow for the direction: AxT hatches HIGHER than wild type, not lower. One plate per genotype at 50% *pos-1* RNAi, a different dose from Figure 4B, so these are not comparable with it and each p describes one plate against one plate.

